01 Cloud Electricity Meter Management

MeterIt Pro is a cloud electricity meter data management platform for meter manufacturers, resellers, utilities and facility teams. Capture readings, validate them, publish reports and query the whole portfolio in plain language — one site or hundreds, from any browser.

24/7 Real-time access to every reading	**100%** Cloud-hosted — no on-prem servers	**15 min** Typical capture interval per register	**RBAC** Eight roles, isolated per tenant

MEASURED AND RETAINED

kWh · kVA · kVA per phase · Amps · Amps per phase · Physical meters · Virtual meters · BACnet · IP

FULL VISIBILITY

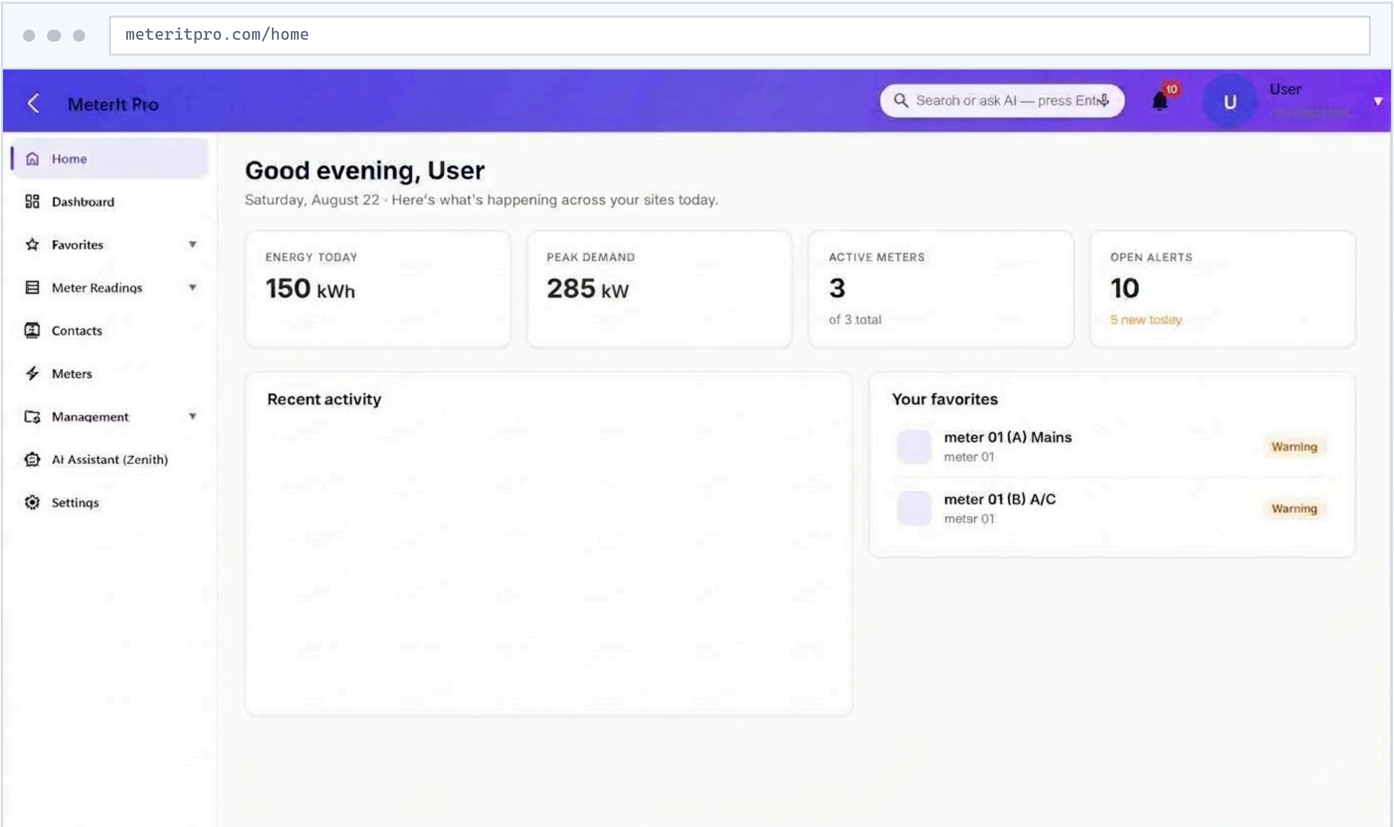
- Live dashboards — no manual export
- Desktop, tablet or phone
- Read-only views for stakeholders

BUILT FOR SCALE

- Hundreds of locations, one portfolio
- Register-level, not building-level
- Scheduled reports by email

SECURE BY DEFAULT

- HTTPS / TLS end to end
- Row-level tenant isolation
- Authentication events logged



Portfolio home — energy today, peak demand, active meters and open alerts at a glance, with recent activity and favourite registers alongside.

02 Core Capabilities

One platform for capture, validation, analytics, alerting and reporting — purpose-built for electricity metering rather than adapted from generic BMS tooling.

Smart meter capture

Mobile-friendly workflows walk technicians through every reading. Physical and virtual meters, captured at the register and circuit level.

Validation & anomaly detection

Quality checks flag anomalies and missing intervals before reports go out, so the data stays audit-ready.

Energy analytics

Usage trends, peak demand and cost efficiency across every meter, on dashboards you compose yourself.

Multi-site portfolio

Readings across hundreds of locations from one dashboard — built for ESCOs, property managers and facility teams.

Alerts & notifications

Threshold, no-reading and zero-reading rules on any register, with in-app and email delivery.

Scheduled reporting

Cost and revenue reports on a cron schedule, delivered as HTML, PDF or CSV with per-recipient delivery logging.

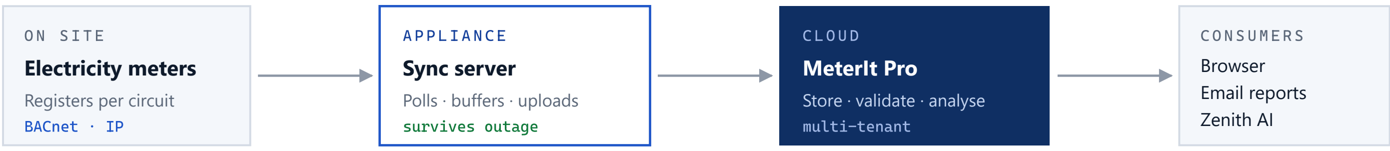
Zenith AI assistant

Plain-language querying over the whole portfolio. Built for AI integration from day one, not retrofitted.

On-site sync server

A sealed Linux appliance that keeps metering through an internet outage and back-fills the cloud on reconnect.

WHAT A DEPLOYMENT LOOKS LIKE



Nothing in the stack above is optional-extra licensing: capture, dashboards, reporting, alerting and the assistant are the same product. The only hardware decision is whether a site needs its own sync-server appliance.

03 Meters & Registers

Every meter is modelled with its own registers, so consumption is tracked at the circuit level rather than the building level. A register is the unit everything else attaches to — readings, graphs, alert rules and favourites.

METER TYPES

Physical meter

Bound to a device on a sync server. MeterIt Pro polls it for live readings on the capture interval.

Virtual meter

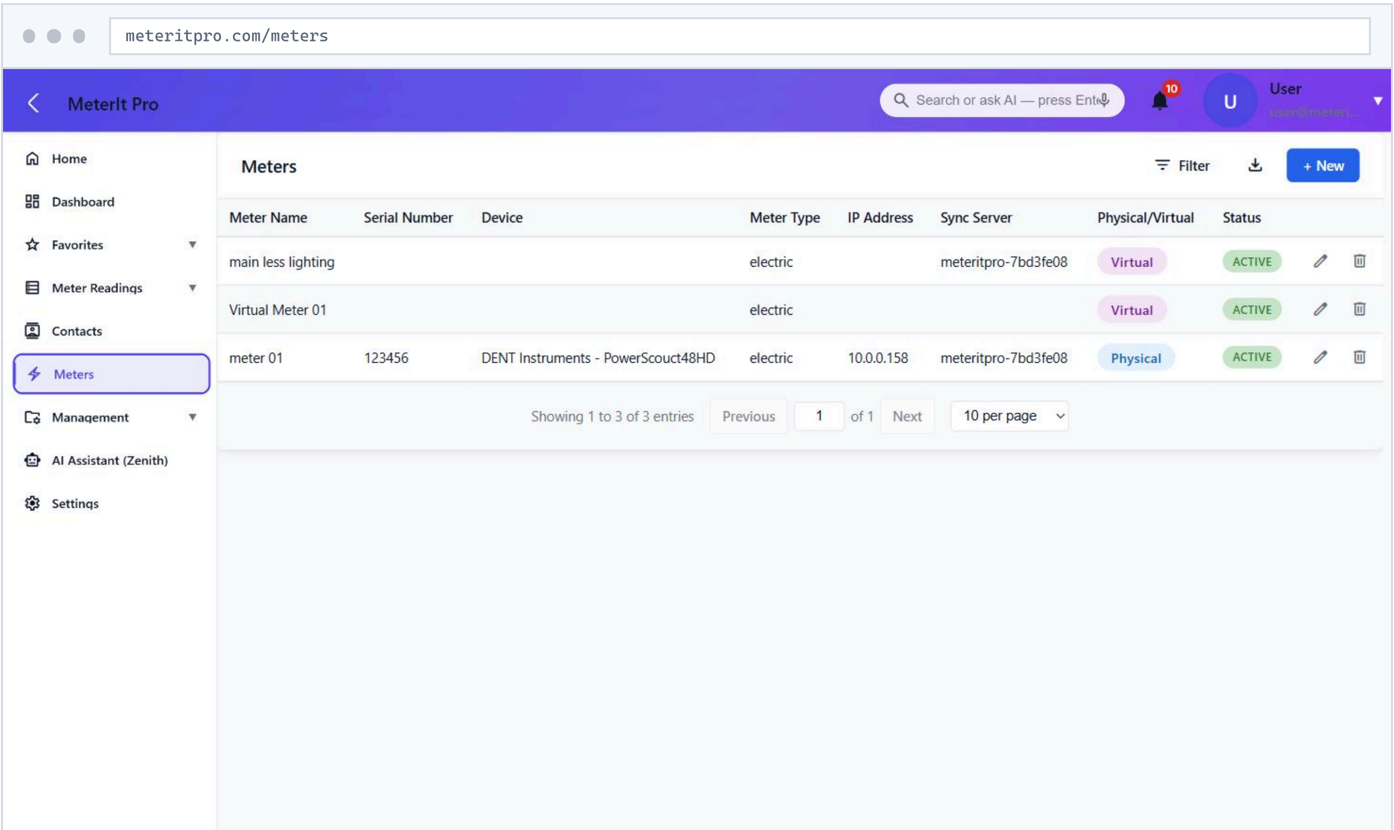
Derived from other meters — mains less lighting, tenant sub-totals, aggregated sites. No hardware, no extra install.

PER-METER REGISTERS

- Multiple registers per meter
- Named channels
- Per-register readings, graphs and alerts
- Serial, IP, sync server and live status

SETUP SEQUENCE

01	Name the meter and assign its location
02	Choose physical or virtual
03	Assign device model and sync server
04	Map registers / elements
05	Set alert rules on any register
06	Assign role-based user access



Meter list — physical and virtual meters side by side, each tied to a device model, IP address, sync server and live status.

04 Metering Data Captured

Per register, per interval — the full electrical picture, timestamped and retained. Every parameter below is queryable, chartable, alertable and export-ready as CSV or PDF.

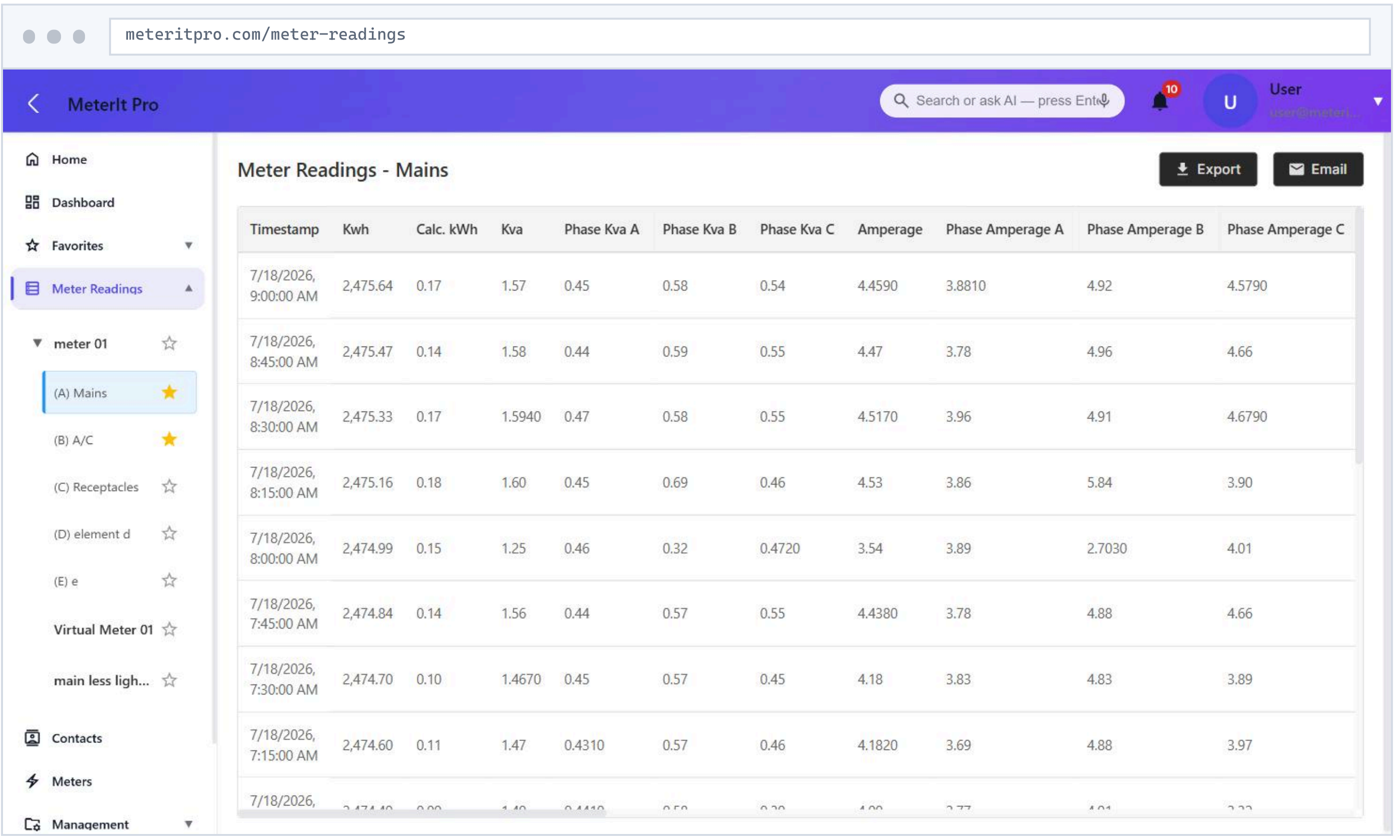
PARAMETER	UNIT	RESOLUTION	NOTES
Energy	kWh	15-min interval	Cumulative and calculated consumption
Apparent power	kVA	15-min interval	Total load
Apparent power, per phase	kVA	15-min interval	Phase A / B / C
Current	A	15-min interval	Total current draw
Current, per phase	A	15-min interval	Phase A / B / C
Timestamp	ISO 8601	per reading	Local site time, retained for audit

GETTING DATA OUT

- **CSV** — raw intervals for billing and analysis
- **PDF** — formatted for audits and clients
- **Email** — one-click send, or on a schedule
- **REST API** — programmatic access per tenant

DATA QUALITY

- Missing intervals surfaced, not silently skipped
- Zero and stalled channels raise their own rule type
- Buffered readings back-fill in correct time order
- Readings are append-only and audit-retained



Timestamp	Kwh	Calc. kWh	Kva	Phase Kva A	Phase Kva B	Phase Kva C	Amperage	Phase Amperage A	Phase Amperage B	Phase Amperage C
7/18/2026, 9:00:00 AM	2,475.64	0.17	1.57	0.45	0.58	0.54	4.4590	3.8810	4.92	4.5790
7/18/2026, 8:45:00 AM	2,475.47	0.14	1.58	0.44	0.59	0.55	4.47	3.78	4.96	4.66
7/18/2026, 8:30:00 AM	2,475.33	0.17	1.5940	0.47	0.58	0.55	4.5170	3.96	4.91	4.6790
7/18/2026, 8:15:00 AM	2,475.16	0.18	1.60	0.45	0.69	0.46	4.53	3.86	5.84	3.90
7/18/2026, 8:00:00 AM	2,474.99	0.15	1.25	0.46	0.32	0.4720	3.54	3.89	2.7030	4.01
7/18/2026, 7:45:00 AM	2,474.84	0.14	1.56	0.44	0.57	0.55	4.4380	3.78	4.88	4.66
7/18/2026, 7:30:00 AM	2,474.70	0.10	1.4670	0.45	0.57	0.45	4.18	3.83	4.83	3.89
7/18/2026, 7:15:00 AM	2,474.60	0.11	1.47	0.4310	0.57	0.46	4.1820	3.69	4.88	3.97
7/18/2026,										

Register readings — kWh, calculated kWh, kVA, per-phase kVA and per-phase amperage at 15-minute intervals, with export and email on the same view.

05 Dashboards & Analytics

Compose dashboards from configurable chart widgets. Range, chart type, granularity and aggregation are set per widget — so each team builds the exact view it needs instead of negotiating over one shared layout.

PER-WIDGET CONTROLS

Range	since install · daily · custom
Granularity	down to 15 min
Aggregation	max · min · avg · sum
Chart type	bar · line · area · pie · table
Actions	refresh · export · email

COMPOSITION

- Multiple dashboards per user
- Drag-and-drop widget layout
- Any register or meter as a series
- Favourites pinned to the portfolio home
- Read-only sharing with stakeholders

Custom dashboard — a Peak kW widget with its range, chart type, granularity and aggregation exposed directly on the widget.

06 Scheduled Reports

Dashboards are for looking; reports are for sending. Define a report once — meters, window, shape, format, recipients — and the platform runs it on a cron schedule and records what actually reached each address.

REPORT DEFINITION

Report type	**cost · revenue**	Schedule	**cron expression**
Time frame	**today · weekly · monthly · yearly · custom**	Grouping	**hourly · daily · weekly · monthly**
Visualization	**bar · line · pie · csv**	Attach as	**html · pdf · csv**
Meter selection	**any meters / registers**	Recipients	**multiple addresses**
State	**active / inactive per report**	Cadence	**recurring, or one-time**

RUN AND DELIVERY PIPELINE

WHY THE DELIVERY LOG MATTERS

- A run that succeeded is not the same as an email that arrived
- Failures keep the error text, per address
- Execution history is indexed by report and by date
- Deleting a report cascades its history and its logs

TYPICAL USES

- Monthly tenant cost recovery, grouped daily
- Weekly revenue summary to the account manager
- Yearly consumption CSV for the auditor
- Custom-window PDF after a tariff change

TWO REPORTS, FULLY SPECIFIED

REPORT	TYPE	WINDOW	GROUPING	SHAPE	GOES OUT AS
Tenant cost recovery active	cost	monthly	daily	bar	pdf
Portfolio revenue summary active	revenue	weekly	daily	line	html

Both are ordinary records — editable, disableable, and duplicable per client. A `csv` visualization with a `csv` attachment produces a raw interval extract instead of a chart.

Reports run server-side on the platform's own schedule, so nothing depends on a browser being open or a workstation being awake. Recipients do not need MeterIt Pro accounts.

07 Alerts & Notifications

Define rules once and the platform watches every register around the clock — so a stalled meter or a runaway load is caught before the client notices it.

RULE TYPES

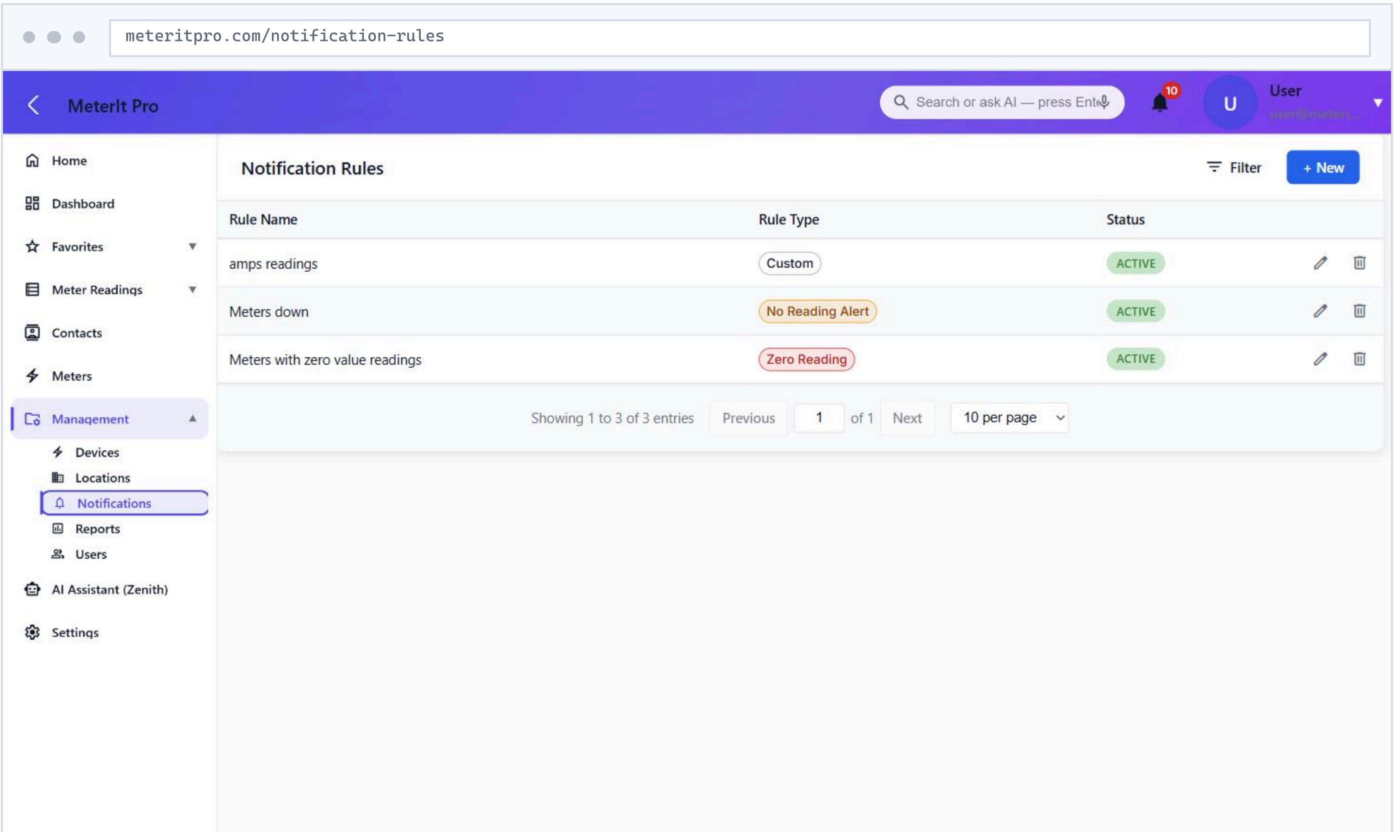
RULE TYPE	FIRES WHEN	WHAT IT CATCHES
Custom `threshold`	A parameter crosses the configured threshold on a register	Runaway load, demand overshoot, tariff breach
No reading `silence`	No reading arrives for a register within the expected interval	Dead comms, offline sync server, decommissioned device
Zero reading `stalled`	A register reports zero where consumption is expected	Dead CT, stalled channel, mis-wired circuit

DELIVERY

- In-app banner on the platform
- Email with a link straight to the meter in alarm
- Threshold breach, over-interval and over-time conditions
- Scheduled scans clear stale notifications

MANAGEMENT

- Active or inactive per rule — no deleting to silence one
- Full notification history retained
- Rules attach to any register, on any meter
- Create, read, update and delete gated by permission



Notification rules — custom, no-reading and zero-reading rule types, each toggled active or inactive per register.

08 Zenith AI Assistant

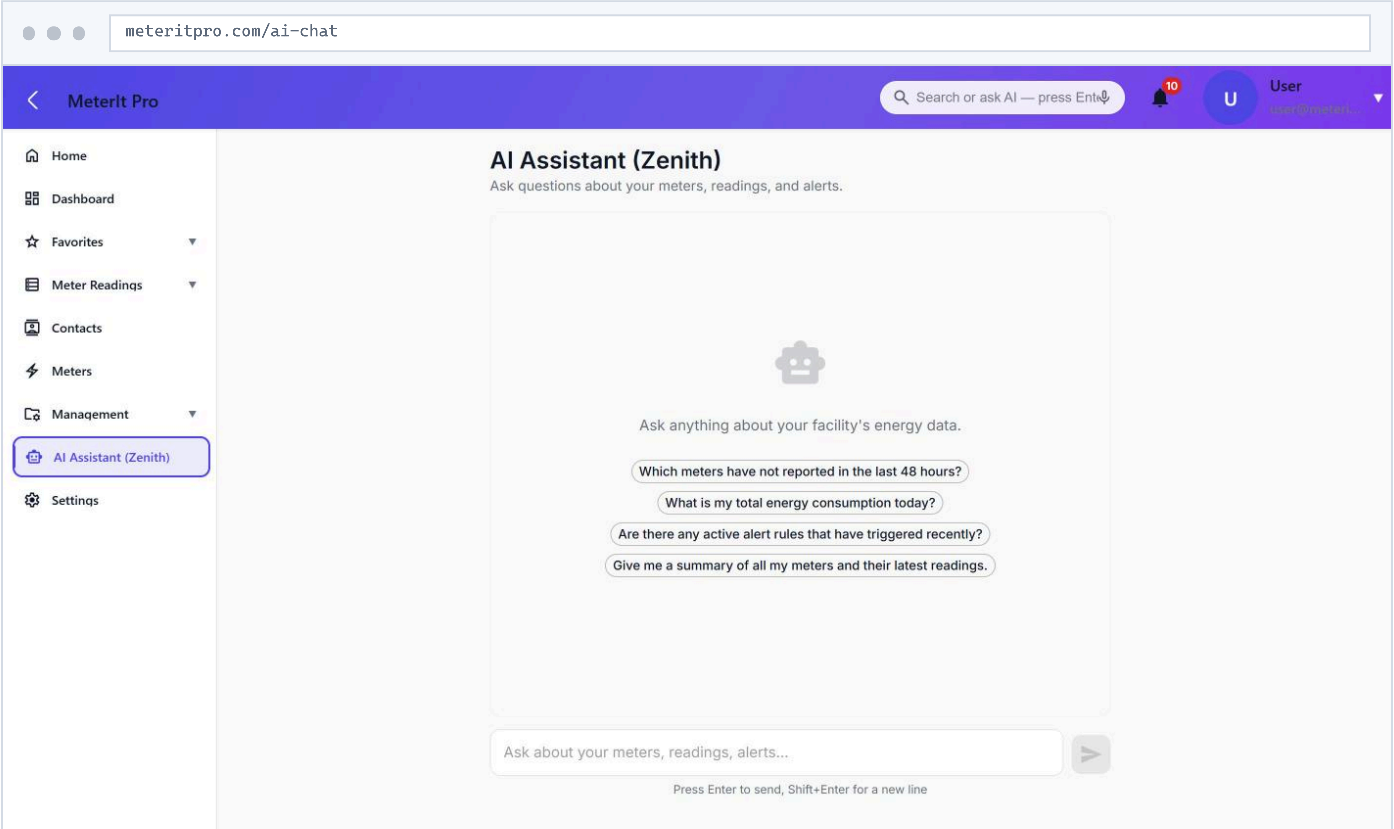
Query meters, readings and alerts in plain language — no report to build, no filters to configure. Zenith reads across the portfolio, inside the asking user's own tenant and permissions, and answers in seconds.

QUESTIONS IT ANSWERS

- Which meters haven't reported in 48 hours?
- What is my total consumption today?
- Which alert rules triggered this week?
- Summarize peak demand across all sites.

HOW IT STAYS TRUSTWORTHY

- Scoped to the caller's tenant — never across tenants
- Reads the same records the UI shows, not a stale copy
- Suggested prompts for a cold start
- Structured data underneath, so answers are checkable



Zenith assistant — a plain-language front door to meter data, with suggested prompts to get started.

09 Locations, Contacts & Organization

A portfolio is not a flat list of meters. Every meter belongs to a location, every location to a tenant, and the people responsible are recorded alongside — so a reading always answers **where** and **whose**, not just how much.

HOW THE RECORDS NEST

TENANT **Company** → LOCATION **Site / building** → METER **Physical or virtual** → REGISTER **Circuit / element**

contacts attach at tenant level and are referenced from any site

LOCATION RECORD

Type	**Warehouse · Apartment · Office**
	Retail · Hotel · Building · Other
Address	**street, unit, city, state, zip, country**
State	**active / inactive**
Attached	**meters, notes, audit stamps**

CONTACT RECORD

Role	**Customer · Client · Vendor**
	Contractor · Technician · Sales Manager
Reach	**email, phone, company**
Address	**full postal address**
State	**active / inactive**

ORGANIZATION SETTINGS — APPLIED ACROSS THE TENANT

Timezone	**IANA zone**	Currency	**per tenant**
Language	**per tenant**	Date format	**e.g. MM/DD/YYYY**
Time format	**12-hour / 24-hour**	Default page size	**rows per list**
Reading batch count	**capture batching**	Identity	**company name, contact email, website**

A PORTFOLIO, AS THE PLATFORM SEES IT

LOCATION	TYPE	METERS	REGISTERS	STATE
Riverside Distribution	Warehouse	4	48	active
Harbour Point Apartments	Apartment	12	36	active
Fifth Street Retail	Retail	3	24	active
Old Mill Office	Office	2	12	inactive

Example figures, shown to illustrate the shape of the records. Deactivating a location keeps its history rather than deleting it.

Timezone and formats are set once per tenant and applied everywhere — reading timestamps, report windows, alert history and exports — so a site in one zone and an accountant in another read the same figures without converting them by hand.

10 Security & Administration

Access is decided in two independent layers: a **role** decides what a person can do, and the **tenant** decides which records exist for them at all. The second layer is enforced in the database, not only in the application.

ROLES

`superadmin` `supersupport` `adminsupport` `admin` `manager` `technician` `viewer` `user`

WHAT EACH ROLE MAY DO

RESOURCE	ADMIN	MANAGER	TECHNICIAN	VIEWER
Meters & registers	create read update delete	create read update	create read update delete	read
Reports	create read update delete	—	create read update delete	—
Notification rules	create read update delete	—	create read update delete	—
Locations	create read update delete	create read update	read	read
Contacts	create read update delete	create read update	read	read
Devices	create read update delete	create read update	read	read
Users	create read update delete	read	read	—
Email templates	create read update delete	read	read	read
Dashboards	create read update delete	create read update delete	read	read
Settings	read update	read	read	read

superadmin matches admin. The two **support** roles and the plain **user** role are read-only, and only support may list users. Each cell is a set of named permissions — `report:create`, `meter:delete` — checked on every API call, so a role can be widened or narrowed without a code change.

ISOLATION & TRANSPORT

- Every record carries its tenant; queries are scoped to it
- Row-level security enforced in PostgreSQL itself
- HTTPS / TLS on every connection
- Rate limiting on authentication endpoints

AUDIT & SUPPORT

- Authentication events logged with status
- IP address and user agent recorded per event
- Created and updated stamps on every record
- In-platform support tickets, admin-triaged

WHAT AN AUTHENTICATION EVENT RECORDS

User	account the event belongs to	Event type	what was attempted
Status	outcome of the attempt	IP address	origin of the request
User agent	client that made it	Timestamp	when it happened

Because tenant isolation lives in the database, a bug in application code cannot leak one customer's meters to another. Role checks then run on top of that, on both the interface and the API — a hidden button is never the only thing standing between a user and a record.

11 On-Site Sync Server

A compact, sealed, **Linux-based** appliance installed on site. It talks to meters over **BACnet and IP** and — the part that matters — **keeps metering when the internet goes down**, then catches the cloud up automatically once the link returns.

RESILIENT BY DESIGN – NO DATA GAPS

ONLINE · NORMAL

Meter & stream live

Polls every meter on the local network at 15-minute intervals and streams readings to the cloud in real time.

INTERNET OUTAGE

Keeps metering offline

If the site loses its link the appliance keeps polling and buffers every reading to local storage. Nothing is lost.

RECONNECTED

Auto-uploads & back-fills

Buffered readings upload automatically and back-fill the timeline in order — no gaps, no manual steps.

SYNC SERVER – DELL MICRO DESKTOP

Front I/O


Rear I/O

ENCLOSURE – ALTELIX 14×11×5 VENTED NEMA

APPLIANCE SPECIFICATIONS

Processor	Intel Core i5-8400T	Cores / clock	6C · 3.3 GHz · 35 W
Memory	16 GB DDR4	Storage	512 GB SSD
Operating system	Linux-based, hardened	Networking	Gigabit Ethernet (RJ-45)
Ports	USB 3.1 · USB-C · 2× DP · HDMI	Form factor	Micro, ≈1.2 L
Dimensions	36 × 182 × 178 mm	Power	65 W external adapter
Meter protocols	BACnet · IP	Local buffering	full readings retained offline
Recovery	fast reimage & redeploy	Uptime	24/7 unattended

ENCLOSURE SPECIFICATIONS

Model	Altelix 14×11×5 PC+ABS vented	Exterior	13.4 × 11.6 × 6.3 in
Interior	12.0 × 8.0 × 4.0 in	Material	Polycarbonate + ABS, UV-resistant
Flame rating	meets UL94-V0	Environmental	NEMA 3R / 3RX · IP24
Door	gasketed hinge, dual latch, key/padlock	Ventilation	2× 3" vents, rain shields, screens
Mounting	wall; optional pole kit to 8" dia.	Weight	3.2 lb (1.4 kg)
RF	transparent for on-site wireless	Included	grommets, ties, ground plate & wire

12 Specifications

Supported hardware and platform characteristics, current at this revision. Additional device models are added on request — the register model is generic, so a new meter is a configuration change rather than a release.

DEVICE COMPATIBILITY

MANUFACTURER	DESCRIPTION	MODEL NO.	TYPE	ELEMENTS
DENT Instruments	PowerScout 48HD	PowerScout48HD	Electric	48
TBWC, Inc.	PS24	DI-MMU8	Electric	24
TBWC, Inc.	PS48	DI-MMU16	Electric	16
TBWC, Inc.	PS12	DI-MMU4	Electric	12
TBWC, Inc.	PS3	DI-SAKIT	Electric	3

PLATFORM

Deployment	**fully cloud-hosted (SaaS)**	On-premises servers	**none required**
Access	**any modern browser**	Tenancy	**multi-tenant, row-level isolated**
Access control	**role-based, 8 roles**	Transport security	**HTTPS / TLS**
Meter focus	**electricity**	Meter models	**physical & virtual**
Reading interval	**15 min typical**	Export formats	**PDF · CSV · HTML · email**
Site sync	**edge sync-server appliance**	Meter protocols	**BACnet · IP**
Scheduling	**cron, server-side**	Integrations	**AI querying · CSV · REST API**

CAPABILITY SUMMARY

- Physical & virtual electricity meters
- Per-register readings, graphs & alerts
- Custom dashboards & analytics
- Scheduled cost & revenue reports
- Multi-site portfolio management
- Threshold, no-reading & zero-reading alerts
- PDF / CSV export & scheduled email
- Zenith AI plain-language querying
- On-site sync server with offline resilience
- Row-level tenant isolation & auth audit log

Enclosure specification source: enclosurehub.com — Altelix 14×11×5 PC+ABS Weatherproof Vented Utility Box. Specifications are subject to change without notice.